BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: Instant Eyes: Subitizing to 5

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it easier to see 4 dots in a row than 4 dots scattered on the floor?'
- Ask: 'How does your brain use the full row of 5 to know what is missing?'
- Ask: 'What happens when you add 1 more to a full row of 5?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students reverting to touching and counting 1, 2, 3, 4. Encourage them to trust their eyes: 'Full row minus 1 is 4!'

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A frame has 3 tokens in the top row and 1 token in the bottom row. What is the total without counting by ones?"
""

OFFICIAL SOLUTION KEY:

3 in top + 1 in bottom = 4 tokens (or 5 minus 1).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: Instant Eyes: Subitizing to 5

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Can your eyes recognize a group of objects instantly without counting them 1, 2, 3? In this lab, train your brain to subitize using the 5-anchor grid.

STEP-BY-STEP APPARATUS SETUP:

1. Place the Ten-Frame Tray in front of you.
2. Notice the top row has exactly 5 circular sockets.
3. Pull 5 tokens (smooth shiny side facing up).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Ten-Frame Tray (2×5 Grid) + 10 Dual-Texture Optical Flip Tokens
DATUM TARGET:	Perceptual Invariant: 5-Frame Anchor Allows Subitizing Without One-by-One Counting
CALCULATOR KEY:	[1] [ENTER] [1] [+] -> 2 (Step Counter)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Kindergarten (KA), Chapter 1 (Numbers to 5)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Kindergarten, Unit 1 (Five-Frames & Counting)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Kindergarten, Ch. 1 • CCSS.MATH.K.CC.B.4 • TEKS K.2A
GRADE LEVEL BAND:	Grades K-1 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 3 ('Counting & Place Value'), pp. 38-52
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 2 (5-Anchor Geometry) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 5-FRAME FLASH GAME

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Place 3 tokens in the top row. Look away. Look back for 1 second. Notice: You instantly see 3!

Step 2: Place 5 tokens in the top row. Notice: A completely full top row ALWAYS means 5 without counting.

Step 3: Partner Flash: Partner A places 1 to 5 tokens, then covers the frame with their hand.

Step 4: Partner A reveals the frame for 1 second, then covers it. Partner B announces the number instantly!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Perceptual Invariant: 5-Frame Anchor Allows Subitizing Without One-by-One Counting

PHYSICAL APPARATUS MANIFEST:

Ten-Frame Tray (2×5 Grid) + 10 Dual-Texture Optical Flip Tok
ens

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: HOW MANY ARE MISSING?

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Look at the empty sockets in the 5-row to find the complement.

FORM A MASTER STATION TASKS (Complete All 3):

1. **If 4 tokens are placed in the top row, how many empty sockets remain?**
   Expected Physical Finding: 1 empty socket (4 + 1 = 5)

2. **If 2 tokens are placed, how many are empty?**
   Expected Physical Finding: 3 empty sockets (2 + 3 = 5)

3. **If 0 tokens are placed, how many are empty?**
   Expected Physical Finding: 5 empty sockets (0 + 5 = 5)

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [ENTER] [1] [+] -> 2 (Step Counter)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Key 1
ENTER	Y: 1 | X: 1	Stack lift
1	Y: 1 | X: 1	Step increment
+	X: 2	Step counter advance

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: Complements of Ten

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do 7 and 3 always belong together when making 10?'
- Ask: 'How does knowing 6 + 4 = 10 help you solve 60 + 40 later on?'
- Ask: 'Why is 5 + 5 the only double fact on the ten-frame?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students losing track of which side is which. The shiny side represents Addend A; the frosted side represents Addend B.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A student holds 10 coins. 7 are heads. How many are tails without counting them one by one?"
""

OFFICIAL SOLUTION KEY:

10 − 7 = 3 tails.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: Complements of Ten

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is 10 the most important number in our counting system? In this lab, flip two-texture tokens to discover all the 'Ten-Friends' that fill the frame.

STEP-BY-STEP APPARATUS SETUP:

1. Fill all 10 sockets of your Ten-Frame Tray with tokens.
2. Start with all 10 tokens showing the shiny specular side (10 + 0 = 10).
3. Prepare to flip tokens one by one to the frosted micro-grooved side.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Ten-Frame Tray + 10 Two-Sided Flip Tokens (Specular Smooth vs Micro-Grooved)
DATUM TARGET:	Conservation of Ten: A + B = 10 (Full 2×5 Frame is Invariant Landmark)
CALCULATOR KEY:	[7] [ENTER] [10] [−] -> 3 (Complement of 7)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 2 (Number Bonds to 10)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 1, Module 3 (Ten-Frames & Making Ten)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 1, Sec. 1.4 • CCSS.MATH.1.OA.C.6 • TEKS 1.3C
GRADE LEVEL BAND:	Grades K–2 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 3, pp. 52–64
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 5 (Two-Color Optical Contrast) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: FLIPPING THE TEN-FRIENDS

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Flip 1 token to frosted: 9 shiny + 1 frosted = 10.

Step 2: Flip another token: 8 shiny + 2 frosted = 10.

Step 3: Flip another token: 7 shiny + 3 frosted = 10.

Step 4: Continue down the line: 6+4, 5+5, 4+6, 3+7, 2+8, 1+9. Notice: The frame is ALWAYS completely full!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Conservation of Ten: A + B = 10 (Full 2×5 Frame is Invariant Landmark)

PHYSICAL APPARATUS MANIFEST:

Ten-Frame Tray + 10 Two-Sided Flip Tokens (Specular Smooth v
s Micro-Grooved)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: BLIND TEN-COMPLEMENT AUDIT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Close your eyes. Partner A flips a secret number of tokens to frosted.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Partner B opens eyes, sees 6 shiny tokens. How many are frosted?**
   Expected Physical Finding: **4 frosted tokens (6 + 4 = 10)**

2. **Partner B sees 2 shiny tokens. How many are frosted?**
   Expected Physical Finding: **8 frosted tokens (2 + 8 = 10)**

3. **Key [10] [ENTER] [6] [−] on SC-6.**
   Expected Physical Finding: **Register X: 4**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [7] [ENTER] [10] [−] -> 3 (Complement of 7)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
10	X: 10	Target frame capacity
ENTER	Y: 10 | X: 10	Stack lift
7	Y: 10 | X: 7	Addend 7
−	X: 3	Extract ten-complement 3

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: Bridging Ten Across the 5mm Rib

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it so much easier to add 10 + 3 than 8 + 5 in your head?'
- Ask: 'How does the 5mm raised divider rib stop your eyes from seeing 12 instead of 10 + 2?'
- Ask: 'Can you use this trick to solve 8 + 7 in 2 seconds?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to fill Frame 2 before Frame 1 is complete. Enforce the Golden Rule: 'Always complete the ten first!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Use bridging to solve 8 + 6. What two numbers do you break 6 into?"
""

OFFICIAL SOLUTION KEY:

Break 6 into 2 and 4. (8 + 2) + 4 = 10 + 4 = 14.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
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https://stackcalc.io/products/ten-frames/#product-videos

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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: Bridging Ten Across the 5mm Rib

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How does mental math work for 8 + 7 or 9 + 5? Stop counting on your fingers: bridge across the 5mm architectural divider to make ten first!

STEP-BY-STEP APPARATUS SETUP:

1. Snap two Ten-Frame Trays together end-to-end using the octagonal MultiBuild connectors.
2. Frame 1 is the 'Tens Anchor'; Frame 2 is the 'Overflow Stage'.
3. Place 8 shiny tokens in Frame 1. Hold 5 frosted tokens in your hand (8 + 5).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Two Ten-Frame Trays (Docked End-to-End via MultiBuild Snaps) + 20 Tokens

DATUM TARGET: Regrouping Invariant: Fill the First Frame to 10 First, Overflow to Second Frame

CALCULATOR KEY: [8] [ENTER] [5] [+] -> 13

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 7 (Addition with Regrouping)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 3 (Double Ten-Frames & Bridging)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 4 • CCSS.MATH.1.OA.C.6 • TEKS 1.3D
GRADE LEVEL BAND:	Grades 1–3 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4 ('Operations & Regrouping'), pp. 68–82
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 2 (5mm Architectural Rib) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE OVERFLOW LOCK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at Frame 1: How many empty sockets are needed to make 10? (Exactly 2 sockets).

Step 2: Take 2 frosted tokens from your hand and lock them into Frame 1. Frame 1 is now FULL (10!).

Step 3: How many frosted tokens remain in your hand? (3 tokens).

Step 4: Place the remaining 3 tokens into Frame 2 across the 5mm divider. Look at the total: 1 full ten + 3 units = 13!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Regrouping Invariant: Fill the First Frame to 10 First, Overflow to Second Frame

PHYSICAL APPARATUS MANIFEST:

Two Ten-Frame Trays (Docked End-to-End via MultiBuild Snaps)
+ 20 Tokens

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE 9-BRIDGE SHORTCUT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Solve 9 + 6 using the bridging strategy.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Place 9 tokens in Frame 1. How many are needed to make 10?**
   Expected Physical Finding: **Only 1 token**

2. **Take 1 from 6 to complete Frame 1. How many remain for Frame 2?**
   Expected Physical Finding: **5 tokens remain (6 – 1 = 5)**

3. **State the total immediately: 1 ten + 5 units.**
   Expected Physical Finding: **15 (9 + 6 = 15)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [8] [ENTER] [5] [+] -> 13

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
8	X: 8	First addend
ENTER	Y: 8 | X: 8	Stack lift
5	Y: 8 | X: 5	Second addend
+	X: 13	Combine and bridge into 1 ten and 3 units

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: The 10-for-1 Base-Ten Regrouping

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can you never have more than 9 loose unit cubes in your answer pile?'
- Ask: 'How does trading 10 cubes for 1 rod make counting faster?'
- Ask: 'What does the little carried 1 at the top of your paper represent in real plastic?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students setting aside 10 cubes without bringing a ten-rod into the total. Always enforce: 'Nothing leaves the bank without a trade!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"You have 3 ten-rods and 15 unit cubes. What number do you actually have after trading?"
""

OFFICIAL SOLUTION KEY:

15 units trades for 1 rod + 5 units. Total = 4 rods and 5 units = 45.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: The 10-for-1 Base-Ten Regrouping

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does carrying a 1 to the next column work in vertical addition? In this lab, you physically trade 10 loose unit cubes for 1 solid ten-rod at the bank.

STEP-BY-STEP APPARATUS SETUP:

1. Place all loose unit cubes on the left ('Ones Bank').
2. Place all solid ten-rods in the center ('Tens Bank').
3. Problem: Add 25 + 18 using physical blocks.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Base-10 Blocks: 20 Unit Cubes (10×10×10mm), 10 Ten-Rods (10×10×100mm)

DATUM TARGET: Exchange Invariant: Exactly 10 Unit Cubes Snap/Align into 1 Ten-Rod (10U = 100mm)

CALCULATOR KEY: [25] [ENTER] [18] [+] -> 43

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2A, Chapter 1 (Numbers to 1,000)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 2 (Place Value & Regrouping)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 5 • CCSS.MATH.2.NBT.A.1 • TEKS 2.2B
GRADE LEVEL BAND:	Grades 1–3 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 3, pp. 64–78
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 2 (10mm Metric Invariant) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE BANK EXCHANGE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Model 25: Take 2 ten-rods and 5 unit cubes.

Step 2: Model 18: Take 1 ten-rod and 8 unit cubes.

Step 3: Combine all loose units: 5 units + 8 units = 13 unit cubes. Notice: You have MORE than 10 loose cubes!

Step 4: The 10-for-1 Trade: Count out 10 loose units and trade them at the bank for 1 solid ten-rod.

Step 5: Count final blocks: 2 + 1 + 1 (new) = 4 ten-rods and 3 remaining units = 43!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Exchange Invariant: Exactly 10 Unit Cubes Snap/Align into 1 Ten-Rod (10U = 100mm)

PHYSICAL APPARATUS MANIFEST:

Base-10 Blocks: 20 Unit Cubes (10×10×10mm), 10 Ten-Rods (10×10×100mm)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: CARRYING TWICE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 47 + 36. Track the unit trade.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Combine units: 7 + 6.**
   Expected Physical Finding: **13 units**
2. **Execute the trade: 10 units -> 1 ten-rod.**
   Expected Physical Finding: **1 new rod, 3 units left**
3. **Combine all rods: 4 + 3 + 1 (traded).**
   Expected Physical Finding: **8 rods**
4. **Final total.**
   Expected Physical Finding: **83**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [25] [ENTER] [18] [+] -> 43

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
25	X: 25	2 rods, 5 units
ENTER	Y: 25 | X: 25	Stack lift
18	Y: 25 | X: 18	1 rod, 8 units
+	X: 43	Regroup: 13 units becomes 1 new rod + 3 units (4 rods, 3 units)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: Decomposing the Ten: Subtraction

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did 52 stay equal to 52 even after we traded a rod for 10 cubes?'
- Ask: 'When you cross out the 5 and write 4 on paper, which block did you move?'
- Ask: 'When you write a little 1 next to the 2 to make 12, where did those 10 cubes come from?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students subtracting smaller from larger in reverse (e.g., doing 7 − 2 = 5 in 52 − 27). Remind them: you must take away from what you STARTED with!

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"Solve 63 − 28 using blocks. What does 63 decompose into?"
""

OFFICIAL SOLUTION KEY:

63 decomposes into 5 rods and 13 units. 13 − 8 = 5 units; 5 − 2 = 3 rods -> 35.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: Decomposing the Ten: Subtraction

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why do we cross out a number and write a smaller number when borrowing in subtraction? In this lab, you physically break open a ten-rod to get more units.

STEP-BY-STEP APPARATUS SETUP:

1. Problem: 52 − 27.
2. Model 52 on your desk: 5 ten-rods and 2 loose unit cubes.
3. Goal: Take away 27 (2 ten-rods and 7 unit cubes).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Base-10 Blocks: 5 Ten-Rods, 20 Unit Cubes + Regrouping Tray

DATUM TARGET: Decomposition Invariant: 1 Ten-Rod Unlocks into Exactly 10 Loose Unit Cubes

CALCULATOR KEY: [52] [ENTER] [27] [−] -> 25

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2A, Chapter 3 (Subtraction with Regrouping)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 3 (Subtraction Strategies)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 6 • CCSS.MATH.2.NBT.B.7 • TEKS 2.4C
GRADE LEVEL BAND:	Grades 2–4 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 82–96
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 6 (Aha! Discovery: Decomposition Proof)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE UNLOCKING STEP

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Try to take away 7 loose units from your pile: You only have 2 loose cubes! You are stuck!

Step 2: The Bank Trade: Pick up 1 of your ten-rods and trade it at the bank for 10 loose unit cubes.

Step 3: Add the 10 cubes to your 2 loose cubes: Now you have 4 ten-rods and 12 loose cubes (40 + 12 = 52).

Step 4: Now take away 7 cubes: 12 − 7 = 5 unit cubes remain.

Step 5: Now take away 2 ten-rods: 4 − 2 = 2 ten-rods remain. Total answer = 25!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Decomposition Invariant: 1 Ten-Rod Unlocks into Exactly 10 Loose Unit Cubes

PHYSICAL APPARATUS MANIFEST:

Base-10 Blocks: 5 Ten-Rods, 20 Unit Cubes + Regrouping Tray

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

--- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ---

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: SUBTRACTING ACROSS ZERO

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 40 − 16. How do you take 6 units from zero units?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Model 40: 4 ten-rods and 0 units.**
   Expected Physical Finding: **Cannot take 6 from 0**

2. **Trade 1 rod for 10 units: What blocks do you hold?**
   Expected Physical Finding: **3 rods and 10 units**

3. **Take away 16 (1 rod and 6 units). What remains?**
   Expected Physical Finding: **2 rods and 4 units = 24**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [52] [ENTER] [27] [−] -> 25

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
52	X: 52	Minuend: 5 rods, 2 units
ENTER	Y: 52 | X: 52	Stack lift
27	Y: 52 | X: 27	Subtrahend: 2 rods, 7 units
−	X: 25	Subtract: 1 rod decomposes into 10 units (12 − 7 = 5 units, 4 − 2 = 2 rods)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Hundreds, Flats & Metric Volume

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'How can the exact same plastic block be 100 on Monday and 0.1 on Friday?'
- Ask: 'Why does moving the decimal point right multiply by 10?'
- Ask: 'How does this connect our base-10 blocks to the Fraction Studio towers?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking decimals are a completely new kind of number. Emphasize: place value simply shifts whether t
he unit is small or large.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Model 2.35 using flats, rods, and units (where Flat = 1, Rod = 0.1, Unit = 0.01). What pieces do you pull?"
""

OFFICIAL SOLUTION KEY:

2 Flats (2.0), 3 Ten-Rods (0.3), and 5 Unit Cubes (0.05).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Hundreds, Flats & Metric Volume

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is our number system metric? A single unit cube is exactly 1 cubic centimeter. Ten of them make a rod, a hundred make a flat, and a thousand make a liter of

STEP-BY-STEP APPARATUS SETUP:

1. Line up 10 ten-rods side-by-side on your desk.
2. Place a Hundred-Flat on top of the 10 rods.
3. Notice: The flat matches the 10 rods in every dimension (100 × 100 mm).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Base-10 Blocks: 2 Hundred-Flats (10×100×100mm), 10 Ten-Rods, 10 Unit Cubes

DATUM TARGET: Metric Proportionality: 1 Cube = 1cm³, 1 Rod = 10cm³, 1 Flat = 100cm³, 1 Large Cube = 1,000cm³ (1 Liter)

CALCULATOR KEY: [345] [ENTER] [278] [+] -> 623

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3A, Chapter 1 (Numbers to 10,000)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 3, Unit 1 (Place Value with 3-Digit Numbers)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 3, Ch. 1 • CCSS.MATH.3.NBT.A.2 • TEKS 3.2A
GRADE LEVEL BAND:	Grades 3–5 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 3, pp. 78–88
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 7 (RPN Register Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE HUNDREDS REGULATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: 10 units = 1 rod (10). 10 rods = 1 flat (100).

Step 2: Stack 10 Hundred-Flats on top of each other: It forms a perfect 1000-cube (100 × 100 × 100 mm).

Step 3: Measure: 1000-cube volume = 1 Liter = 1,000 grams of water (1 kilogram).

Step 4: Conclude: In base-10, every place value step is a 10× multiplication in physical space.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Metric Proportionality: 1 Cube = 1cm³, 1 Rod = 10cm³, 1 Flat = 100cm³, 1 Large Cube = 1,000cm³ (1 Liter)

PHYSICAL APPARATUS MANIFEST:

Base-10 Blocks: 2 Hundred-Flats (10×100×100mm), 10 Ten-Rods,
10 Unit Cubes

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: DECIMAL REINTERPRETATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Declare the large 1000-cube to be '1 WHOLE'.

FORM A MASTER STATION TASKS (Complete All 3):

1 **If the 1000-cube is 1 WHOLE, what does a Hundred-Flat represent?**
Expected Physical Finding: **0.1 (One Tenth)**

2 **What does a Ten-Rod represent?**
Expected Physical Finding: **0.01 (One Hundredth)**

3 **What does a single Unit Cube represent?**
Expected Physical Finding: **0.001 (One Thousandth)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [345] [ENTER] [278] [+] –> 623

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
345	X: 345	3 flats, 4 rods, 5 units
ENTER	Y: 345 | X: 345	Stack lift
278	Y: 345 | X: 278	2 flats, 7 rods, 8 units
+	X: 623	Regroup both columns: 13 units –> 1 rod, 12 rods –> 1 flat (6 flats, 2 rods, 3 units)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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